Applied Soil Ecology 217 (2026) 106570

Contents lists available at ScienceDirect

Applied Soil Ecology

journal homepage: www.elsevier.com/locate/apsoil

Vermicompost-mediated suppression of root-knot nematodes relates to tomato root bacterial community composition under greenhouse conditions

Chen Liang [a,1], Leixin Yu [b,1], Qinghai Yang [c], Lijuan Yang [a,*], Chuang Yang [d,*], Xuguang Niu [a], Feiyu Dong [a], Diwen Yang [a]

[a] Land and Environmental College, Shenyang Agricultural University, Shenyang, Liaoning, 110866, PR China
[b] Shenyang Urban Construction University, Shenyang, Liaoning, 110167, PR China
[c] Eastern Liaoning University, Dandong, Liaoning, 118003, PR China
[d] Shenyang Hengxin Technology Management Consulting Service Co., Ltd, Shenyang, Liaoning, 110000, PR China

ARTICLE INFO

Keywords:
Root-knot nematodes
Vermicompost
Tomato
Bacterial community
Bacillus
Pseudomonas

ABSTRACT

Root-knot nematodes (RKN) poses a significant threat to global vegetable production, exhibiting a high incidence. Organic fertilizers are rich in microorganisms and effective against soil-borne diseases. This study investigated the effects of vermicompost on soil chemical properties, rhizosphere and endophytic bacterial communities of tomato roots infected with *Meloidogyne incognita* (*M. incognita*). The results indicated that vermicompost increased soil pH, ammonium and nitrate levels. Additionally, the ratio of soil carbon to nitrogen (C/N) and soil microbial biomass also increased with a higher proportion of vermicompost. When the proportion of vermicompost reached 80 %, the number of root knots and the root index of tomatoes were reduced by 59.91 % and 60.54 %, respectively, compared with no fertilizer. Vermicompost significantly enhanced the microbial population and the ratio of bacteria to fungi in the soil, while also altering the composition and diversity of the tomato root system. In the rhizosphere of tomato plants, the abundance of *Bacillus* increased with the application of vermicompost. In the roots of tomatoes, *Pseudomonas* also increased with the application of vermicompost, ultimately becoming the dominant bacterium. The structural model demonstrated that *Bacillus* and *Pseudomonas* had significant inhibitory effects on RKN. Soil nutrients ($NH4^{+}$-N, C/N) and pH exhibited indirect inhibitory effects on RKN. Furthermore, *Bacillus* and *Pseudomonas* were positively correlated with the aboveground biomass of tomatoes. This study contributes to our understanding of the regulatory effects of vermicompost on the microecosystem of RKN-infected tomato roots and provides a scientific basis for the sustainable production of vegetables and the ecological management of RKN.

1. Introduction

In recent years, continuous cropping has become a significant bottleneck obstructing the progress of the vegetable sector, garnering widespread attention (Alami et al., 2021). Long-term continuous cropping will lead to the disruption of soil microbial ecological balance, microbial flora pathologies, and soil-borne disease intensification (Sun et al., 2023). The root-knot nematodes is among the most prevalent and damaging soil-borne pathogens under continuous cropping of vegetables in facilities (Coyne et al., 2018). Its high incidence, large disaster area and difficult prevention often lead to significant crop reduction or failure (Burns et al., 2023). At present, the main measures to control RKN are mainly chemical agents, but their high toxicity and high resides destroy the ecological environment, harm the health of humans and animals, and most nematocides are gradually banned or strictly restricted. In the process of vegetable production, it is an urgent problem to seek safe and efficient biological control measures against RKN (Zuhair et al., 2022). Activities based on agriculture and animal husbandry produce millions of tons of organic and inorganic solid waste, which intensifies the global environmental degradation. In terms of sustainable agriculture, the conversion of agricultural waste into reusable resources is critical. Many kinds of organic waste, including green manure, livestock manure and straw (Anedo et al., 2024), have become a research hotspot because of their wide sources, simple and easy to

* Corresponding authors.
E-mail addresses: lijuanyangsyau@126.com (L. Yang), ycydylx@163.com (C. Yang).
[1] These authors contributed equally to this work.

https://doi.org/10.1016/j.apsoil.2025.106570
Received 8 March 2025; Received in revised form 10 October 2025; Accepted 24 October 2025
Available online 4 November 2025
0929-1393/© 2025 Elsevier B.V. All rights reserved, including those for text and data mining, AI training, and similar technologies.

obtain, improving soil environment, preventing soil-borne diseases and optimizing plant growth. Vermicompost results from the biodegradation of organic waste by earthworms in conjunction with diverse microorganisms, which can greatly improve the microbial quantity and microbial activity in soil (Chew et al., 2019; Rehman et al., 2023). It contains some antagonistic microorganisms and can synthesize a series of bioactive substances to enhance the disease resistance of agricultural products (Song et al., 2021). The application of vermicompost has been shown in recent studies to inhibit particular pathogenic fungi such as *Rhizoctonia solani*, *Botrytis cinerea* (Nakasone et al., 1999), *Fusarium oxysporum* (Zhao et al., 2019), *Phytophthora cryptogea* (Orlikowski, 1999), *Phytophthora nicotianae* (Szczech and Smolińska, 2001). In addition, the application of vermicompost can also inhibit the occurrence of diseases such as aphid (*Myzus persicae*), cabbage white caterpillar (*Peiris brassicae*) and yellow mealworm (*Pseudococcus* spp.) (Arancon et al., 2002).

Previous studies have confirmed that vermicompost has significant inhibitory effects on plant parasitic nematodes (Edwards et al., 2007), and its mechanism needs further research and exploration. The possible mechanism of inhibition includes improving soil health and plant nourishment, enhancing plant disease tolerance, containing nematocidal compounds or releasing nematocidal substances, increasing or introducing microorganisms antagonizing RKN, improving the number and diversity of rhizosphere microorganisms, and indirectly stimulating nematode predators and parasites dependent on microbial activity. Xiao et al. (2016) found that vermicompost can effectively improve the physical structure and chemical composition of soil, regulate the defense response of tomato roots and leaves, and improve the nematode resistance. Mishra et al. (2018) used root splitting test and real-time fluorescent quantitative PCR method and proved that the vermicompost extract induced resistance of cucumber plants to *M. incognita* and reduced the penetration of nematodes to cucumber roots by upregulating the expression of plant defense genes. Rostami et al. (2021) screened and isolated *Sphingobacterium daejeonense* from vermicompost, which showed high nematocidal potential in tomatoes infected with *Meloidogyne javanica*. Studies have shown that if plants are infected with diseases, the original ecological balance of rhizosphere microbial communities will be broken and the microbial flora will become pathological (Yang et al., 2017; Zhou et al., 2020). The effects of RKN on rhizosphere microbial communities of tobacco (Lu et al., 2023), soybean (Barros et al., 2022), eggplant (Zhou et al., 2019a, 2019b), tomato (Tian et al., 2015) and cucumber (Yang et al., 2023) have been reported one after another. This demonstrates that rhizosphere microorganisms are closely related to the severity of plant root-knot nematode disease. Bacteria, as the most abundant organisms in soil, are widely distributed and exhibit broad-spectrum insecticidal activity, showing great potential for application in the biological control of nematodes (Meyer, 2003; Sikora et al., 2007). Among them, rhizosphere bacteria represent an important group against RKN (Hackenberg et al., 2000; EL-Nagdi and Youssef, 2004). Vermicompost contain abundant microorganisms that can increase soil bacterial biomass, directly introduce beneficial bacteria into the soil, improve soil environment, and promote the proliferation of beneficial bacteria (Zhao et al., 2019). Currently, there are few reports on the regulation of vermicompost on the rhizosphere microbial community structure of RKN-infected soil.

As a new biological control agent, vermicompost serves a crucial function in the treatment of solid organic waste, agricultural production and ecological environment protection. This study evaluated the control effect of different application ratios of vermicompost on *M. incognita* in tomato. High-throughput sequencing technology was used to analyze the changes of bacterial community structure and diversity in tomato roots, explore the regulatory effects of vermicompost on tomato "rhizosphere - endorhizosphere" microecosystem, predict the functional microorganisms and key factors of vermicompost inhibiting RKN, and further explore the prevention and control mechanism of vermicompost on tomato infested by RKN.

2. Materials and methods

2.1. Experimental site and soil characteristics

The experiment was carried out in the greenhouse of the Scientific Research Station at Shenyang Agricultural University. Soil infected with RKN was sourced from a tomato-cultivated greenhouse of Chaoyang, Liaoning province, China. Because of the long-term continuous cropping, RKN disease seriously affected tomato production, and the yield showed a trend of decreasing. The initial population density of nematodes was determined to be 1095.0 ± 26.5 J2s/100 g soil using the Baermann funnel method (Arshad et al., 2023; Jenkins, 1964). Healthy soil without RKN contamination was also obtained from the same region for comparison. Surface soils (0–20 cm depth) from RKN-infected and healthy plots were air-dried at 25 °C for 72 h, manually crushed to break aggregates, and passed through a 5-mm sieve to remove coarse debris. Subsequently, soils were pre-incubated for 10 days at 25 °C and 60 % water-holding capacity in darkness to stabilize microbial activity and minimize transient fluctuations in nutrient availability. Moisture levels were monitored daily and adjusted with sterile distilled water. Fresh vermicompost was produced by feeding *Eisenia foetida* earthworms with partially decomposed cattle manure. After 3 months, the non-worm material was separated by 2-mm sieve and earthworm eggs and impurities were removed. The remaining materials with particles less than 2-mm were vermicompost. The basic properties of the original soil and vermicompost are shown in Table 1.

2.2. Treatments and sample collection

In the experiment of potted tomato seedlings, 7 treatments with different ratio (mass ratio) of vermicompost to infested soil were set up: (1) 100 % infested soil without vermicompost (VM0); (2) 20 % vermicompost +80 % infested soil (VM1); (3) 40 % vermicompost +60 % infested soil (VM2); (4) 60 % vermicompost +40 % infested soil (VM3); (5) 80 % vermicompost +20 % infested soil (VM4); (6) 100 % vermicompost (VM5); (7) Healthy soil without vermicompost (CK). Each treatment was repeated 5 times. Tomato seedlings with 5 leaves in consistent growth condition were selected and planted in a 15 cm × 13 cm pot. The potted plants were transplanted randomly and wetted timely and quantitatively. The whole growth period was 45 days with no additional fertilization.

Forty-five days after transplantation, soil samples were collected and removed debris for soil chemical property determination. The growth characteristics and disease grade in tomato plants were measured and analyzed. 10 g of tomato rhizosphere soil was sampled according to the shaking technique (Liang et al., 2014), which was used to determine microorganisms in the soil. In addition, rhizosphere soil, root knot and non-root knot parts of tomato roots were taken and placed in sterile plastic bags and frozen at −80 °C for total genomic DNA extraction.

2.3. Assessment of soil chemical properties and microbial characteristics

The pH of the soil was measured with a PHSJ-3F digital acidity meter after treating with CO_2-removed distilled water at a soil-to-water ratio of 2.5:1. Soil NO_3^--N was extracted using calcium chloride and determined by ultraviolet spectrophotometer. Soil NH_4^+-N was extracted with potassium chloride and determined by indophenol blue colorimetry. Soil total carbon (TC) and total nitrogen (TN) were quantified through elemental analysis methods. Soil moisture content was assessed using the drying method. Microbial biomass carbon (SMC) and nitrogen (SMN) were measured through chloroform fumigation followed by potassium sulfate extraction.

2.4. Evaluation of tomato growth and diseases incidence

After harvesting the roots of the tomato plants, the soil on the root

Table 1
Basic properties of soils and vermicompost used in the greenhouse experiment.

Materials	pH	NO_3^--N (mg· kg^{-1})	NH_4^+-N (mg· kg^{-1})	TN (g· kg^{-1})	OM (g·kg^{-1})	C/N
Healthy soil	6.84 ± 0.04	27.16 ± 1.15	4.63 ± 0.09	1.36 ± 0.08	20.85 ± 0.72	8.89 ± 0.08
Infected soil	6.14 ± 0.06	23.67 ± 1.05	3.26 ± 0.06	1.26 ± 0.12	15.77 ± 1.02	7.26 ± 0.05
Vermicompost	7.15 ± 0.10	33.24 ± 1.62	3.77 ± 0.05	11.35 ± 1.03	258.70 ± 4.06	13.28 ± 0.24

NO_3^--N: nitrate nitrogen; NH_4^+-N: ammonium nitrogen; TN: Total nitrogen; OM: organic matter; C/N: carbon nitrogen ratio.

surfaces was gently washed, and sucked up with absorbent paper. The height of the plant, stem diameter, aboveground and belowground biomass, as well as the number of root-knots on tomato roots were all measured. According to the grade standard of RKN disease, the disease level was recorded (Barker, 1985), and the root-knot index (RKI) were calculated.

$$\text{RKI}\,(\%) = \Sigma\,(\text{Number of diseased plants at each level} \times \text{Corresponding level value})/(\text{Total number of plants surveyed} \times \text{Highest Level}) \times 100\%$$

2.5. Determination of soil microbial quantity

The population of microorganisms was determined by the dilution plate method (Li et al., 2008). Beef extract peptone agar medium, Martin's medium, and modified Gao's I medium were used for the isolation of bacteria, fungi, and actinomycetes, respectively.

2.6. DNA extraction, amplification and microbial community analysis

Given the close interaction between root-knot nematodes and plant rhizosphere bacteria as well as endophytic bacteria (Zuckerman et al., 1989; Ferrara et al., 2012). this study primarily investigated the changes in the structure and diversity of plant root-associated bacterial communities. Genomic DNA from plant roots and rhizosphere soil was extracted using the FastDNA® Spin Kit for Soil (MP Biomedicals, USA) in accordance with the extraction protocol. The concentration and purity of the genomic DNA were determined by the NanoDrop 2000 ultraviolet-visible spectrophotometer (Thermo Fisher Scientific, USA), and DNA quality was confirmed by 2 % agarose gel electrophoresis. The purified DNA samples were amplified with primers 799F (5'-AACMGGATTAGATACCCKG-3') and 1193R (5'-ACGTCATCCCCACCTTCC-3'). The PCR reaction mixture was prepared as follows: 5 × FastPfu Buffer 4 μL, 2.5 mmol/L dNTPs 4 μL, forward primer 799F (5 μM) 0.8 μL, reverse primer (5 um) 0.8 μL, *FastPfu* polymerase 0.4 μL, BSA 0.2 μL, template DNA 10 ng, ddH_2O supplemented to 20 μL. The PCR reaction conditions were: pre-denaturation at 95 °C for 3 min, 27 cycles (denaturation at 95 °C for 30s, annealing at 55 °C for 30s, extension at 72 °C for 30s), and then stable extension at 72 °C for 10 min. The PCR amplicons were isolated using a 2 % agarose gel, subsequently purified with a DNA Gel Recovery and Purification Kit (PCR Cleanup Kit, Over China), and their concentration was measured using a Qubit 4.0 fluorometer (Thermo Fisher Scientific, USA). The final sequencing was conducted on the Illumina PE250 platform (Shanghai Meiji Biomedical Technology Co., Ltd.).

2.7. Bioinformatics and statistical analyses

Sequence analysis is conducted using UPARSE (v7.0.1001, UPARSE OTU clustering). Based on a 97 % similarity threshold, OTUs (Operational Taxonomic Units) were employed to cluster the quality-controlled concatenated sequences and remove chimeric sequences. To minimize the influence of sequencing depth on subsequent *alpha* and *beta* diversity analyses, all sample sequences were normalized. The taxonomic classification of OTUs was carried out using the RDP Classifier (version 2.11) in conjunction with the Silva 16S rRNA gene database (v138), with a confidence threshold set at 70 %. The microbial community composition of each sample was analyzed and categorized at various taxonomic levels.

The *alpha* diversity indices, including Ace, Chao, and Shannon, were computed using the mothur software (http://www.mothur.org/wiki/Calculators). Statistical differences in *alpha* diversity between groups were assessed using the Wilcoxon rank-sum test. Principal coordinate analysis (PCoA) employing the Bray-Curtis distance metric was utilized to evaluate the similarity of microbial community structures among samples. Additionally, a non-parametric permutation test (PERMANOVA) was applied to determine whether significant differences existed in the microbial community composition between samples. The UPGMA algorithm was used to perform cluster analysis on different community distance matrices, and the sample-level cluster tree was constructed to visually show the similarity or difference degree of community composition in different samples. In order to screen out the microorganisms that have significant influence on RKN, Random Forest analysis was performed on major microorganisms at genus level by the "randomForest" package in R software (vsesion 3.3.1). Structural equation models (SEM) were constructed by Spasspro online platform (https://www.spsspro.com/?utm_source=baidu-ss) to examine the direct and indirect influences of soil environment and plant characteristics on RKN. The model goodness-of-fit was evaluated using multiple indices, including the chi-square statistic (χ^2), the goodness-of-fit index (GFI), and the comparative fit index (CFI). The stability and significance of the path coefficients were assessed using a non-parametric bootstrap procedure with 5000 iterations.

Statistical analyses of plant growth and disease incidence data were conducted using one-way analysis of variance (ANOVA) in SPSS 22.0. Prior to ANOVA, the normality of all datasets was confirmed using the Shapiro-Wilk test ($p > 0.05$), and the homogeneity of variances was verified for most datasets using Levene's test ($p > 0.05$). For the few datasets where Levene's test indicated a violation of variance homogeneity ($p < 0.05$), Welch's ANOVA was applied. All post-hoc comparisons were conducted using Duncan's new multiple range test at a significance level of $P < 0.05$.

3. Results

3.1. Vermicompost alleviates RKN damage and improves soil chemical environment

To investigate the potential of vermicompost to control RKN, we first evaluated its effects on tomato health and soil properties. As shown in Table 2, compared to the healthy soil, the RKN-infested soil severely inhibited the aboveground growth of tomato seedlings, while the underground biomass abnormally increased due to gall (root knots) formation. Application of vermicompost significantly restored the aboveground biomass of tomatoes, which peaked when the vermicompost proportion reached 80 % (VM4). Concurrently, the belowground dry weight was the lowest in the VM4 treatment, being 43.02 % lower than that in the VM0 treatment. Furthermore, vermicompost application significantly reduced both the number of root knots and the root-knot index (RKI), with the VM4 treatment showing the most prominent disease suppression effect and the lowest RKI (27.62 %).

This growth recovery and disease suppression were closely associated with the restorative effects of vermicompost on the infested soil. As shown in Fig. 1 (A) and (B), vermicompost effectively reversed the soil

Table 2
Effects of vermicompost on the biomass and *M. incognita* of tomato seedlings.

Treatments	Aboveground dry weight (g)	Underground dry weight (g)	Number of galls	Root knot index (%)
CK	5.82 ± 0.77d	0.63 ± 0.05bc	–	–
VM0	3.18 ± 0.31e	0.86 ± 0.03a	230.40 ± 2.03a	70.00 ± 0.77a
VM1	9.63 ± 0.79c	0.84 ± 0.01ab	158.00 ± 5.69b	47.27 ± 1.70b
VM2	12.56 ± 0.50b	0.84 ± 0.13ab	153.00 ± 1.76b	45.77 ± 0.52b
VM3	12.24 ± 0.54b	0.81 ± 0.09ab	137.00 ± 22.52b	40.98 ± 6.74b
VM4	18.09 ± 1.27a	0.49 ± 0.02c	92.40 ± 3.18c	27.62 ± 0.95c
VM5	6.63 ± 0.13d	0.63 ± 0.00bc	–	–

CK: Healthy soil; VM0: 100 % infested soil; VM1: 20 % vermicompost +80 % infested soil; VM2: 40 % vermicompost +60 % infested soil; VM3: 60 % vermicompost +40 % infested soil; VM4: 80 % vermicompost +20 % infested soil; VM5:100 % vermicompost. Different letters in the same column indicate significant difference at $p < 0.05$ by Duncan's test.

acidification and decreased C/N ratio induced by RKN infection. Soil pH and C/N ratio generally exhibited an increasing trend with higher vermicompost application rates. Simultaneously, vermicompost application significantly enhanced soil NH_4^+-N and NO_3^--N contents. When the application ratio of vermicompost was between 40 % and 80 %, soil NH_4^+-N were significantly increased under VM4, VM3 and VM2 treatment, corresponding to 28.58 %, 14.84 % and 14.79 % higher than CK treatment, respectively (Fig. 1 C). Except VM1 treatment, soil NO_3^--N in other treatments was significantly higher than VM0 treatment (Fig. 1 D).

Additionally, The application of vermicompost increased SMBC and SMBN, while decreasing the microbial biomass carbon-to-nitrogen ratio (SMBC:SMBN), indicating a shift in soil microbial activity and community structure (Fig. 2 A-C).

3.2. Vermicompost reshapes the structure and diversity of rhizospheric and endophytic bacterial communities

Given the potential role of soil microorganisms in disease suppression, we further analyzed the response of the bacterial community in the soil. It can be observed that the addition of vermicompost significantly increased the total abundance of soil microorganisms. With increasing vermicompost application rates, the populations of soil bacteria and fungi, as well as the bacteria-to-fungus ratio, also increased (Table S1). More importantly, diversity analysis indicated that vermicompost treatments significantly enhanced the richness (Ace and Chao index) and diversity (Shannon index) of bacterial communities in both the rhizosphere and root endosphere (Supplementary Fig. S1). Principal coordinate analysis (P CoA) and hierarchical clustering analysis revealed that vermicompost treatments altered the bacterial community structure, showing a clear gradient relationship with the application rate (Fig. 3A and B).

We further analyzed the composition of bacteria communities at phylum and genus taxonomic ranks under different treatments. At the phylum level, the predominant bacterial groups across all treatments were Proteobacteria, Firmicutes, among others (Fig. 3 C). At the genus

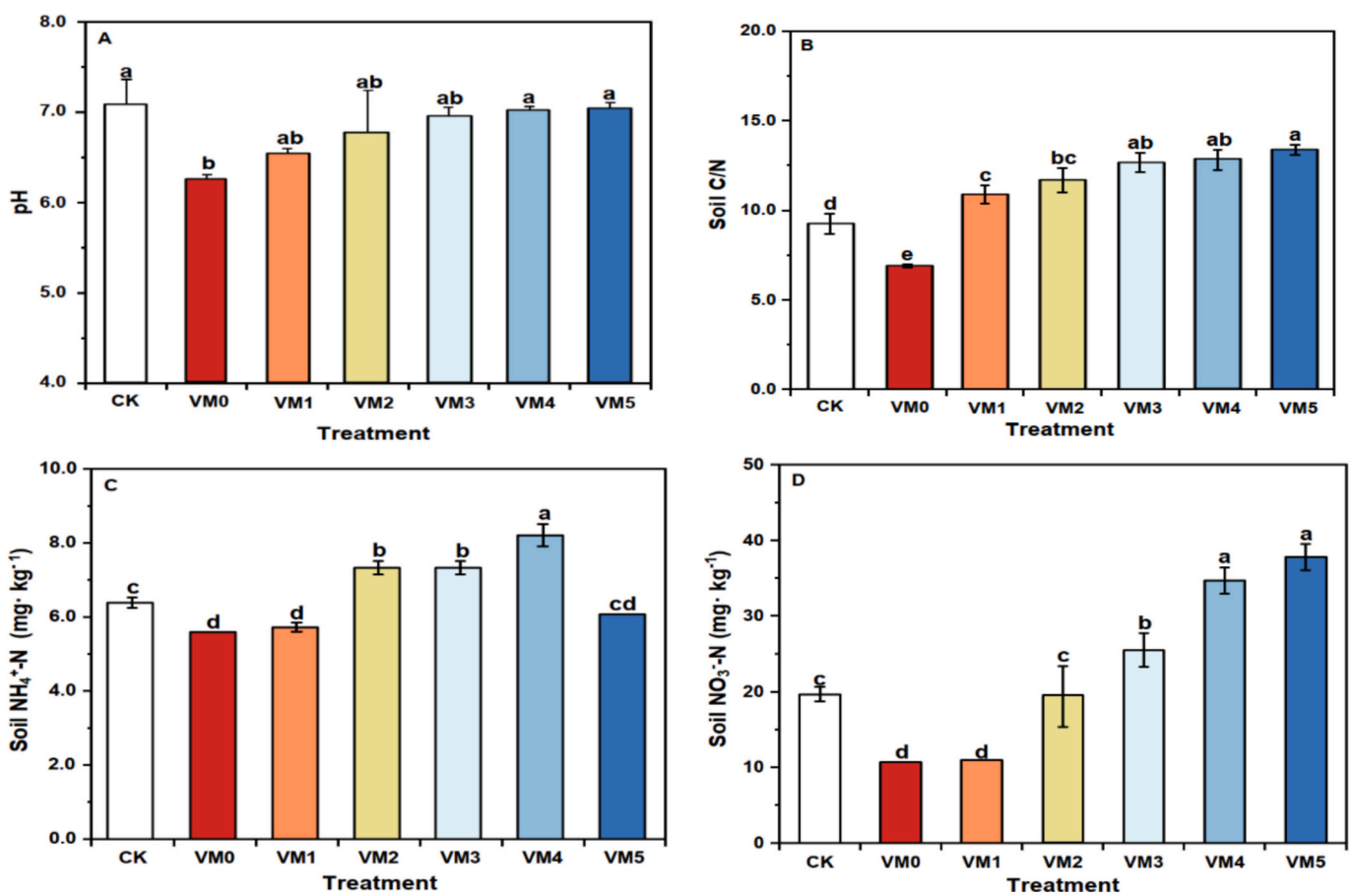

Fig. 1. Effects of vermicompost on soil chemical properties. A. Determination of pH in Soil. B. Soil Carbon to nitrogen ratio, Soil C/N. C. Soil ammonium nitrogen content, Soil NH_4^+-N. D. Soil nitrate nitrogen content, Soil NO_3^--N. CK: Healthy soil; VM0: 100 % infested soil; VM1: 20 % vermicompost +80 % infested soil; VM2: 40 % vermicompost +60 % infested soil; VM3: 60 % vermicompost +40 % infested soil; VM4: 80 % vermicompost +20 % infested soil; VM5:100 % vermicompost. Different letters in the same column indicate significant difference at $p < 0.05$ by Duncan's test. Bars indicate the standard error of the means ($n = 5$).

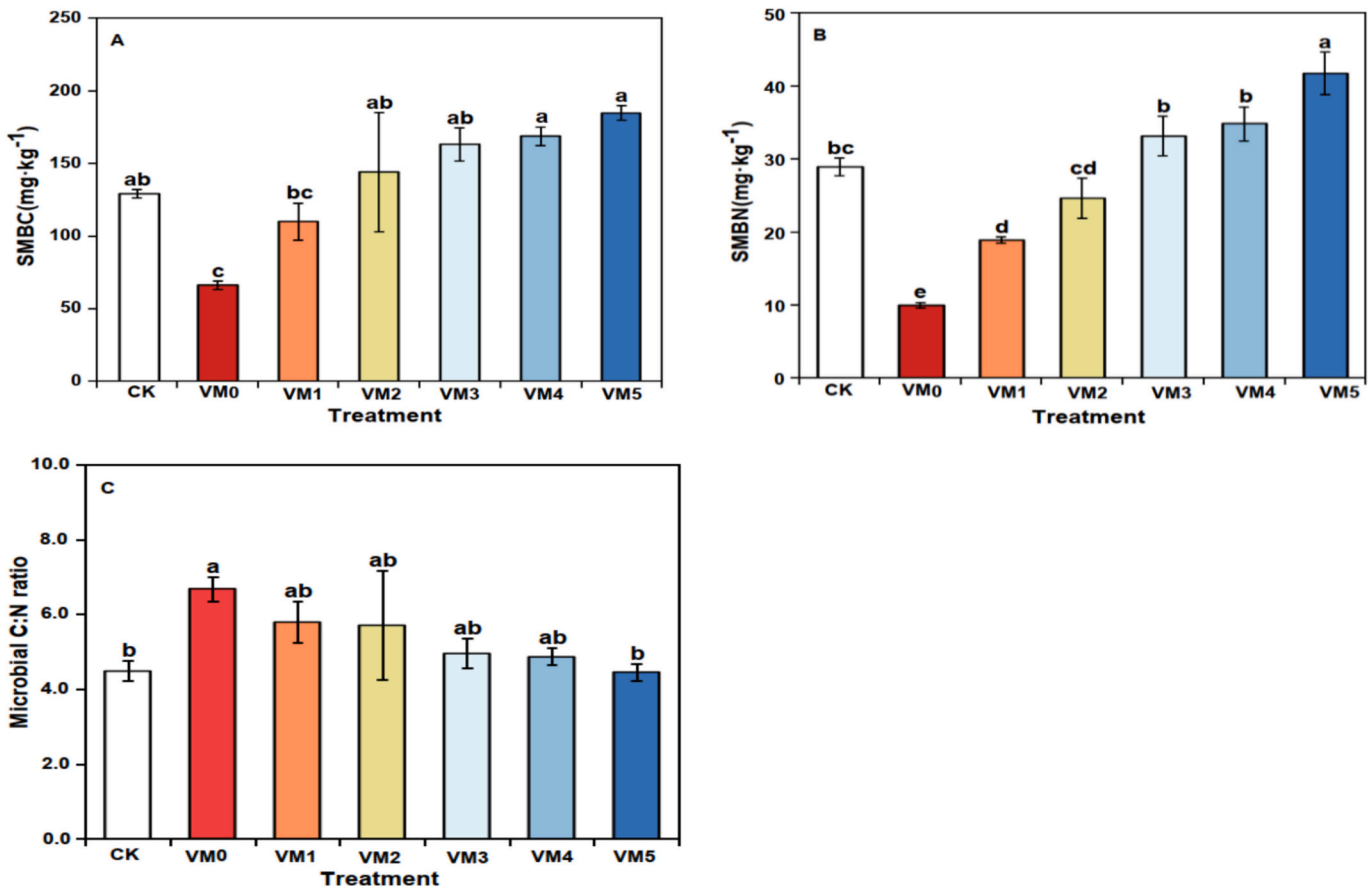

Fig. 2. Effects of vermicompost on soil microbial biomass. A. Soil microbial biomass carbon content, SMBC. B. Soil microbial biomass nitrogen content, SMBN. C. Microbial biomass carbon to nitrogen ratio, Microbial C/N ratio. CK: Healthy soil; VM0: 100 % infested soil; VM1: 20 % vermicompost +80 % infested soil; VM2: 40 % vermicompost +60 % infested soil; VM3: 60 % vermicompost +40 % infested soil; VM4: 80 % vermicompost +20 % infested soil; VM5:100 % vermicompost. Different letters in the same column indicate significant difference at $p < 0.05$ by Duncan's test. Bars indicate the standard error of the means ($n = 5$).

level, we observed significant shifts in key taxa. In the rhizosphere, *Bacillus* was significantly enriched following vermicompost application. Within the root system, *Pseudomonas* became the dominant genus in the treatments, particularly in the non-root-knot parts of VM4 (Fig. 3D). These findings indicated that the disease-suppressive effect of vermicompost is closely associated with its driving role in the succession of core microbial taxa.

3.3. Key bacterial taxa associated with vermicompost-induced RKN suppression in tomato

To precisely identify the key members with disease-suppressive functions from the successional microbial communities, we conducted random forest analysis (Fig. 4 A and B). The random forest model predicted that the key bacteria associated with disease suppression in both the rhizosphere and root endosphere were primarily distributed among the phyla Firmicutes, Proteobacteria, and Actinobacteria. The dominant bacterial genera identified in tomato rhizosphere soil through genus-level analysis included *Bacillus*, *Sporosarcina*, *Lysinibacillus*, *Pseudorhodoplanes*, and *Gaiella*. In the tomato root system, the main bacterial genera were *Lysinomonas*, *Shinella*, *Flavobacterium*, *Azoarcus* and *Hydrogenophaga*. Most of these bacteria were relatively low in abundance, belonging to relatively rare bacterial groups.

Spearman's correlation analysis further revealed direct links between these key taxa and plant health or disease severity. In the rhizosphere system, the relative abundances of *Bacillus* and *Planifilum* were negatively correlated with the Shannon index ($r = -0.73$, $***p < 0.001$; $r = -0.55$, $**p < 0.01$) and RKI ($r = -0.60$, $**p < 0.01$; $r = -0.60$, $***p < 0.001$) (Fig. 4 C). In the tomato root system, the abundance of *Pseudomonas* exhibited a negative correlation with the Ace index ($r = -0.42$, $*p < 0.05$) and RKI ($r = -0.45$, $**p < 0.01$) (Fig. 4 D).

We subsequently tracked the dynamic changes of these key bacterial genera within the rhizosphere and root system under various treatments. It was found that the relative abundance of *Bacillus* in rhizosphere soil was continuously enriched with increasing vermicompost application rates (Fig. 5 A). Meanwhile, within the roots system, the relative abundance of *Pseudomonas* progressively accumulated in the non-root-knot parts of tomato plants as the amount of vermicompost application increased (Fig. 5 B). This strongly suggests that *Bacillus* and *Pseudomonas* play central roles in the vermicompost-mediated suppression of RKN.

3.4. Integrated model reveals the pathway of disease suppression mechanism

We integrated the above findings through structural equation modeling (SEM), which systematically elucidated the overall pathway of "soil environment - key microorganisms - disease suppression - plant health" (Fig. 6). The model results clearly demonstrated that vermicompost indirectly inhibits the root-knot index (RKI) by enhancing factors such as soil pH and C/N ratio. The key bacteria, *Bacillus* and *Pseudomonas*, not only exerted direct and significant negative effects on RKI, but *Bacillus* also significantly facilitated the accumulation of *Pseudomonas*. Concurrently, both bacterial taxa had significant positive effects on the aboveground biomass of tomato plants.

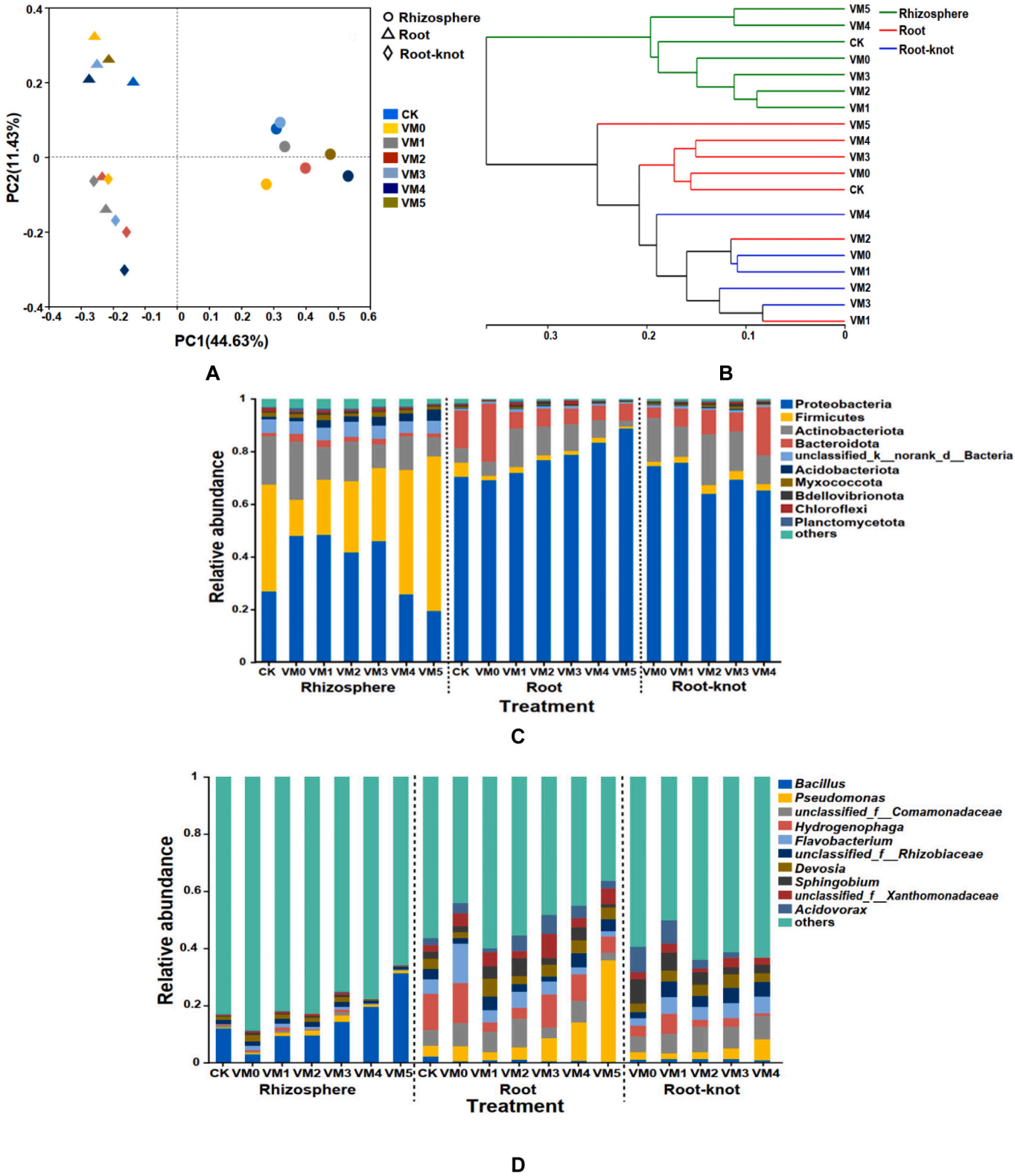

Fig. 3. Effects of vermicompost application on microbial community composition: Principal coordinate analysis (PCoA, A), UPGMA-based hierarchical clustering (B), and relative abundance at the phylum (C) and genus (D) levels.CK: Healthy soil; VM0:100 % infested soil; VM1: 20 % vermicompost +80 % infested soil; VM2: 40 % vermicompost +60 % infested soil; VM3: 60 % vermicompost +40 % infested soil; VM4: 80 % vermicompost+20 % infested soil; VM5: 100 % vermicompost.

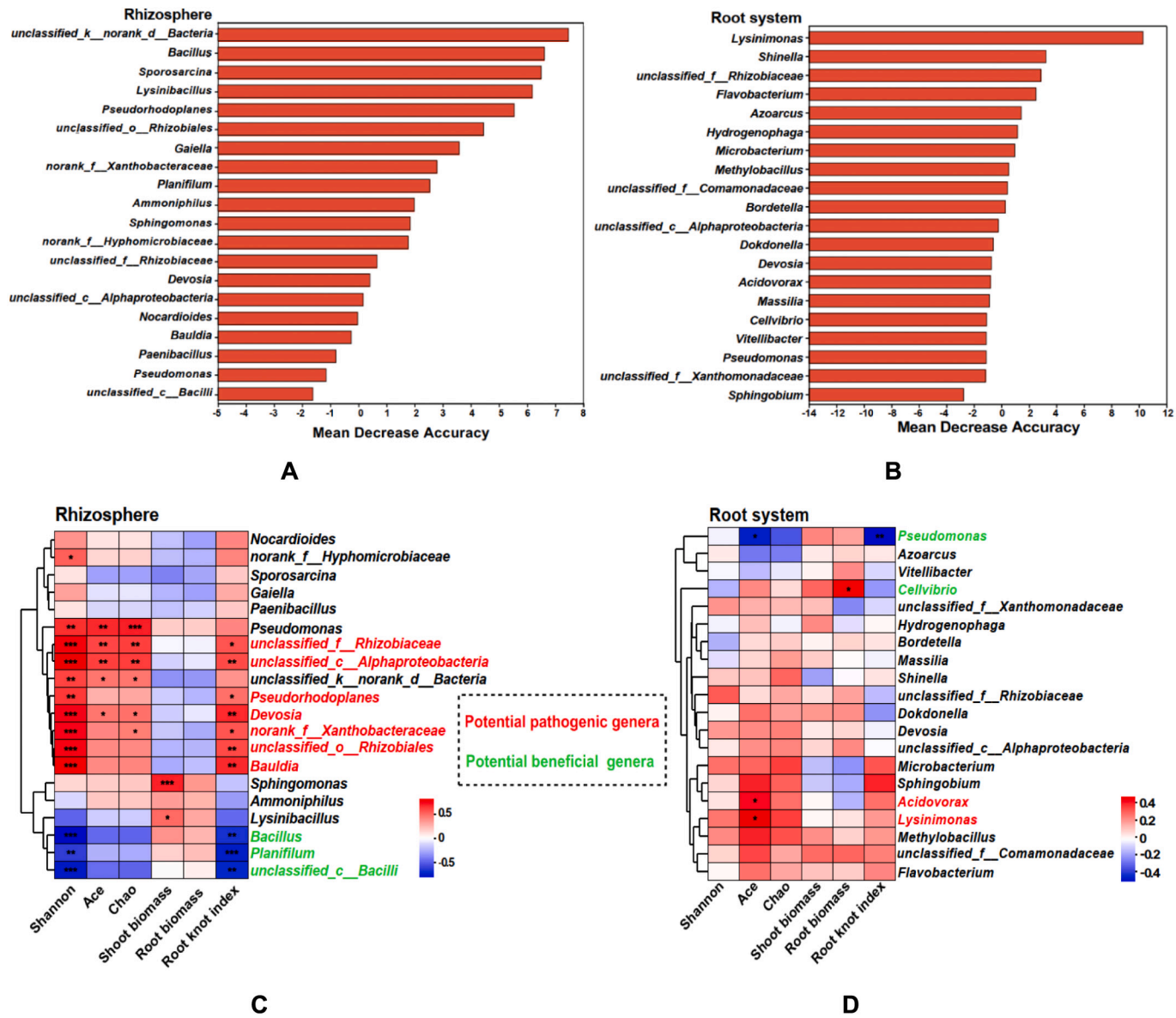

Fig. 4. Main different microorganisms at the genus level (A, B) and heat map analysis between potential microbial taxa and plant growth characteristics and disease incidence (C, D). Spearman correlation coefficient corresponding to each taxon showed positive and negative correlation at *0.05, ** 0.01, ***0.001 significance level.

4. Discussion

4.1. Vermicompost suppresses RKN disease and promotes tomato growth through multiple synergistic pathways

This study confirms that vermicompost application effectively alleviates the suppressive effect of RKN on tomato growth, significantly reduces the root-knot index, with an 80 % application rate showing the optimal efficacy. This finding aligns with the suppression of *Meloidogyne incognita* by vermicompost reported by Liu et al. (2017), which indicated that tomato resistance to RKN could be significantly enhanced when the vermicompost proportion exceeds 75 %. However, this 45-day experiment was conducted under controlled greenhouse conditions, which may not fully reflect field environmental complexities or account for economic feasibility. Field studies suggested vermicompost can be recommended as an effective biocontrol agent at application rates of 10–20 t ha^{-1} for effective nematode suppression (Gebrehana et al., 2025; Zuhair et al., 2022).

The disease suppression and growth promotion effects of vermicompost stem from its comprehensive improvement of the soil ecosystem. Primarily, as a high-quality organic fertilizer, it significantly enhances soil pH, C/N ratio, and nitrogen availability (Fig. 1), directly ameliorating the soil chemical environment deteriorated by RKN infestation. More importantly, the SEM (Fig. 6) reveals that the improvements in soil pH and C/N ratio do not directly affect the nematodes but indirectly exert their suppressive function by promoting the enrichment of key disease-suppressive microorganisms. The increase in soil pH creates favorable conditions for disease-suppressive bacteria preferring neutral-alkaline environments, such as *Bacillus* (Wang et al., 2025). Meanwhile, the elevated C/N ratio and NH_4^+-N provide ample nutrients and energy for microbial activity, potentially suppressing nematode activity by promoting ammonia release or direct nutrient competition (Oka, 2010; Lazarovits et al., 2001). Therefore, vermicompost initially lays the foundation for the subsequent microbial pathogen suppression process by improving the soil conditions.

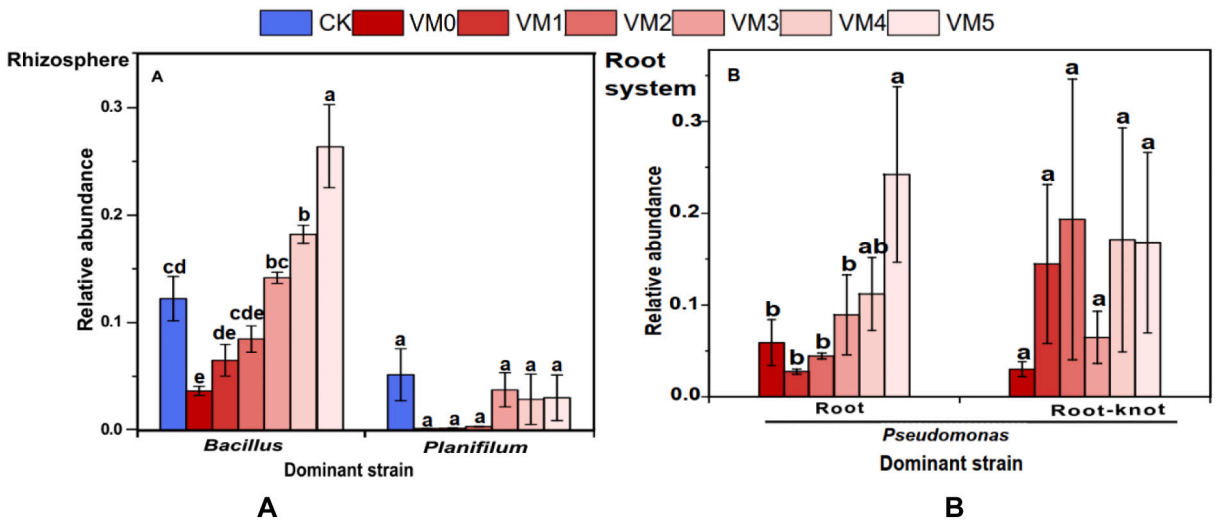

Fig. 5. Relative abundance of major functional bacteria in rhizosphere (A) and root (B) with different amounts of vermicompost. CK: Healthy soil; VM0:100 % infected soil; VM1: 20 % vermicompost +80 % infected soil; VM2: 40 % vermicompost +60 % infected soil; VM3: 60 % vermicompost +40 % infected soil; VM4: 80 % vermicompost +20 % infected soil; VM5: 100 % vermicompost. Different letters in the same column indicate significant difference at $p < 0.05$ by Duncan's test. Bars indicate the standard error of the means ($n = 5$).

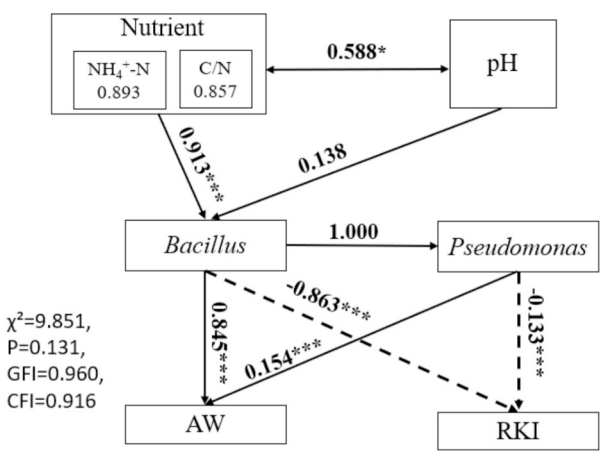

Fig. 6. Structural equation model based on the influence of tomato rhizosphere and root system related indexes. The number next to the arrow represents the path coefficient; "*,**,***" respectively represents significant level in 1 %、5 %、10 %; AW: Aboveground biomass; RKI: Root knot index; Solid lines represents positive correlations, and dotted lines represents negative correlations.

4.2. Microbial community restructuring is a key link in vermicompost functionality

Continuous cropping is an important restriction factor for large-scale and intensive production of facilities. Long-term continuous cropping will lead to the disruption of soil microbial ecological balance, the pathologies of microflora, and the aggravation of soil-borne diseases. RKN represent one of the most devastating soil-borne pathogens affecting tomato cultivation worldwide. Emerging research indicated a strong correlation between plant diseases and biodiversity of microbial ecosystems. The community structure and interactions between rhizosphere and endophytic microorganisms in plants may largely determine the severity of disease development. Therefore, it is of great significance to explore the changes of rhizosphere and root microecological environment during the infection of tomato plants by RKN for the growth, development and yield quality of crops (Tian et al., 2015).

Our results confirmed that RKN infection reduces the diversity of bacterial communities in both the rhizosphere and root endosphere of tomatoes (Supplementary Fig. S1), aligning with the findings of Zhang et al. (2020) in potatoes, where the chao and shannon indices of rhizosphere bacteria in potatoes infected with RKN were significantly lower than in healthy plants. This demonstrates that nematode invasion indeed disrupts the stability of the root-associated microecology. The advancement of this study lies in our identification of vermicompost as an effective means to reverse this damage and reshape a healthy microecology. The addition of vermicompost not only comprehensively enhanced microbial abundance and diversity but, more importantly, drove a directional succession of the community structure (Fig. 3). The core feature of this succession is the coordinated enrichment of specific beneficial bacterial genera in different ecological niches of the rhizosphere and root endosphere. Our data clearly show that vermicompost significantly enriched *Bacillus* in the rhizosphere and *Pseudomonas* in the root endosphere. These two genera are well-recognized biocontrol agents, whose disease-suppressive mechanisms have been extensively reported, and commercial biocontrol products have been developed based on them (Siddiqui, 2004; Dong and Zhang, 2006; Wahla et al., 2012). Szczech (1999) previously indicated that the disease-suppressive effect of vermicompost relies on its bioactive components. Our findings provide an interpretation of this phenomenon from a community ecology perspective: vermicompost, as a highly bioactive soil amendment, utilizes both its inherent microorganisms and the soil environment it creates to collectively select and support these indigenous microorganisms with disease-suppressive functions, thereby reprogramming the soil microbial community into a disease-suppressive state.

4.3. Enrichment and synergy of key disease-suppressive microorganisms are core to vermicompost efficacy

Through random forest and correlation analyses, this study precisely identified *Bacillus* in the rhizosphere and *Pseudomonas* in the root endosphere as two core functional taxa from the complex microbial community reshaped by vermicompost. Their relative abundances showed significant negative correlations with the root-knot index and increased with higher vermicompost application rates (Figs. 4, 5). Extensive studies have identified several *Bacillus* as effective RKN antagonists, including *Bacillus firmus* (Settu et al., 2023), *Bacillus thuringiensis* (Chen et al., 2023), *Bacillus pumilus* (Dai et al., 2023), *Bacillus amyloliquefaciens* (Gowda et al., 2022), *Bacillus altitudinis* (Ye et al., 2022), *Bacillus subtilis* (Dinardo-Miranda et al., 2022), *Bacillus cereus*

(Yin et al., 2021), *Bacillus simplex* (Wang et al., 2016) and *Bacillus megaterium* (Antil et al., 2022). Similarly, multiple *Pseudomonas* demonstrated antagonistic effects against RKNs, such as *Pseudomonas fluorescens* (Siddiqui et al., 2005), *Pseudomonas protegens* (Li et al., 2015), *Pseudomonas chlororaphis* (Lee et al., 2011), *Pseudomonas aeruginosa* (Siddiqui and Shaukat, 2003) and *Pseudomonas putida* (Zhai et al., 2018). These bacteria employed diverse mechanisms to control RKNs, including production of nematicidal toxins (Cry proteins), secretion of chitinases and proteases degrading nematode eggshells and cuticles, induction of plant systemic resistance (ISR), and nutrient and niche competition (Siddiqui, 2004; Tian et al., 2007). Through comprehensive screening, we successfully isolated three *Bacillus* strains with antagonistic activity from vermicompost, demonstrating over 80 % suppression efficacy against nematodes in tomato plants. These strains produced key active substances like proteases, and chitinases, capable of degrading or penetrating nematode eggshells, thereby disrupting normal egg development and damaging the morphological structure of J2, ultimately leading to ovicidal and nematicidal effects (Liang et al., 2024).

In this study, we not only verified the enrichment of *Bacillus* and *Pseudomonas* in vermicompost-amended soil but also made a key discovery: the SEM revealed a synergistic relationship between them, where rhizospheric *Bacillus* promoted the accumulation of root-endophytic *Pseudomonas* (Fig. 6). This finding elevates the understanding of the suppression mechanism from single-strain functions to the level of "microbial collaboration" (Yu et al., 2024; Fan et al., 2025). We hypothesize that *Bacillus*, as a pioneer colonizer in the rhizosphere, may provide more readily available nutrients for *Pseudomonas* or alter root exudate composition through its metabolic activities, thereby facilitating the migration and colonization of the latter within the roots. This speculation aligns with conclusions by Sun et al. (2021) regarding interactions among beneficial rhizosphere bacteria. Furthermore, studies have indicated that modulating microbial community structure via metabolic products is a common strategy for *Bacillus* (Xia et al., 2022). This cross-niche microbial collaboration establishes multiple disease-suppressive barriers from the rhizosphere (first line of defense) to the root interior (second line of defense), significantly enhancing the stability and efficiency of disease suppression. This provides an important theoretical basis for the future design and development of highly efficient, ecologically rational composite microbial inoculants.

4.4. An integrated disease-suppression model: from soil amendment to microbial collaboration

The structural equation model (SEM) developed in this study provides a comprehensive and mechanistic understanding of how vermicompost regulates the soil-plant-microbe system to suppress root-knot nematodes (Fig. 6). The model quantitatively confirmed a primary pathway: the suppression of RKN by vermicompost is an indirect process, predominantly mediated by the succession of the soil microbial community. This finding aligns with the current view that the disease-suppressive effects of organic amendments primarily occur through biological pathways rather than chemical means (Zhao et al., 2019; Tao et al., 2020). Specifically, the increases in soil pH and C/N ratio created a favorable soil physicochemical environment, thereby promoting the enrichment of key bacterial taxa.

Our model revealed a critical finding: a clear driving relationship exists between these key taxa, namely, *Bacillus* in the rhizosphere exhibited a significant positive effect on the accumulation of *Pseudomonas* within the roots. This reveals a cross-niche microbial synergy, whereby the rhizosphere colonizer likely paves the way for, or facilitates, the root colonization of the endophytic antagonist. Consequently, the enriched *Bacillus* and *Pseudomonas*, acting as direct executors of nematode suppression, both showed significant negative effects on the root-knot index. This directional driving effect of the rhizosphere bacterium on the endophytic bacterium provides new evidence for understanding how soil microbial communities construct multi-level, ordered defense networks. Ultimately, the successful suppression of RKN was directly translated into improved plant growth performance, as evidenced by the significant positive effects of these two bacteria on aboveground biomass. This integrated model systematically elucidates the internal mechanism by which vermicompost regulates the soil micro-ecosystem to enhance tomato resistance against RKN, providing a solid theoretical foundation for employing vermicompost as an ecology-based plant protection strategy.

5. Conclusion

Our study demonstrates that vermicompost amendments can effectively suppress root-knot nematodes (RKN) in tomatoes through a process primarily mediated by the restructuring of root-associated microbial communities. The key innovative finding of this research is the identification of a synergistic relationship between specific bacterial taxa across different root compartments: vermicompost application consistently enriches *Bacillus* in the rhizosphere, which in turn promotes the accumulation of the antagonistic bacterium *Pseudomonas* within the root endosphere. Structural equation modeling (SEM) confirmed that nematode suppression is achieved not through direct chemical effects, but indirectly via this orchestrated microbial synergy. This interaction between Bacillus and Pseudomonas across root compartments represents a crucial mechanism underlying the biocontrol effect. Therefore, our work provides a novel, mechanistic understanding of vermicompost-induced disease suppression and offers a theoretical foundation for developing targeted microbial consortia or organic amendments for the sustainable management of root-knot nematodes.

This study primarily validated the effects and mechanisms of vermicompost under greenhouse conditions. Although the SEM revealed a potential synergy between *Bacillus* and *Pseudomonas*, the specific molecular interaction mechanisms require further elucidation through axenic systems, metagenomic, metatranscriptomic studies, and other approaches. Future research should transition to validation in complex field environments and incorporate economic considerations, which will be crucial for advancing the practical application of vermicompost.

CRediT authorship contribution statement

Chen Liang: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Leixin Yu:** Writing – review & editing, Software, Conceptualization. **Qinghai Yang:** Methodology, Investigation. **Lijuan Yang:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization. **Chuang Yang:** Visualization, Validation, Supervision. **Xuguang Niu:** Writing – original draft, Project administration, Conceptualization. **Feiyu Dong:** Investigation, Conceptualization. **Diwen Yang:** Writing – review & editing, Data curation.

Funding

This research was financially supported by the Program of Distinguished Professor of Liaoning Province China (01062920001).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors thank all those who provided helpful suggestions on this manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apsoil.2025.106570.

Data availability

Sequencing data are deposited in NCBI Sequence Read (SRA) database as SUB15459056 (accession numbers: XXXXXX). All other data will be made available on request.

References

Alami, M.M., Pang, Q.L., Gong, Z.D., Yang, T.W., Tu, D.Q., Zhen, O.Y., Yu, W.L., Alami, M.J., Wang, X.K., 2021. Continuous cropping changes the composition and diversity of bacterial communities: a meta - analysis in nine different fields with different plant cultivation. Agriculture 11, 1224.

Anedo, E.O., Beesigamukama, D., Mochoge, B., 2024. Evolving dynamics of insect frass fertilizer for sustainable nematode management and potato production. Front. Plant Sci. 15, 1–17.

Antil, S., Kumar, R., Pathak, D.V., Kumar, A., Panwar, A., Kumari, A., 2022. Plant growth-promoting rhizobacteria-*Bacillus cereus* KMT-5 and *B. megaterium* KMT-8 effectively suppressed Meloidogyne javanica infection. Appl. Soil Ecol. 174, 1–11.

Arancon, N.Q., Edwards, C.A., Lee, S.S., Yardim, E.Y., Bolumu, B.K., Turkey, V., 2002. Management of plant parasitic nematode populations by use of vermicomposts. Manag. Plant Parasit. Nematode 1–6.

Arshad, K., Amir, K., Asgar, A., Saba, F., Ahmad, S.M., 2023. Root-knot nematodes (*Meloidogyne* spp.): biology, plant-nematode interactions and their environmentally benign management strategies. Gesunde Pflanz. 75, 2187–2205.

Barker, K.R., 1985. Design of greenhouse and microplot experiments for evaluation of plant resistance to nematodes. In: Zuckerman, B., Mai, W.F., Harrison, M.B. (Eds.), Plant Nematology Laboratory Manual. Food and Agriculture Organization of the United Nations, Rome, pp. 103–113.

Barros, F.M.D., Pedrinho, A., Mendes, L.W., Freitas, C.C.G., Andreote, F.D., 2022. Interactions between soil bacterial diversity and plant - parasitic nematodes in soybean plants. Appl. Environ. Microbiol. 88, 1–14.

Burns, A.R., Baker, R.J., Kitner, M., Knox, J., Cooke, B., Volpatti, J.R., Vaidya, A.S., Puumala, E., Palmeira, B.M., Redman, E.M., Snider, J., Marwah, S., Chung, S.W., MacDonald, M.H., Tiefenbach, J., Hu, C., Xiao, Q., Finney, C.A.M., Krause, H.M., MacParland, S.A., Stagljar, I., Gilleard, J.S., Cowen, L.E., Meyer, S.L.F., Cutler, S.R., Dowling, J.J., Lautens, M., Zasada, I., Roy, P.J., 2023. Selective control of parasitic nematodes using bioactivated nematicides. Nature 618, 1476–4687.

Chen, F., Pang, C.Y., Zheng, Z., Zhou, W., Guo, Z., Xiao, D., Du, H., Bravo, A., Soberón, M., Sun, M., 2023. Aminopeptidase MNP-1 triggers intestine protease production by activating daf-16 nuclear location to degrade pore-forming toxins in *Caenorhabditis elegans*. PLoS Pathog. 19, 1–27.

Chew, K.W., Chia, S.R., Yen, H.W., Nomanbhay, S., Ho, Y.C., Show, P.L., 2019. Transformation of biomass waste into sustainable organic fertilizers. Sustainability 11, 1–19.

Coyne, D.L., Laura, C., Dalzell, J.J., Claudius-Cole, A.O., Solveig, H., Nessie, L., Herbert, T., 2018. Plant-parasitic nematodes and food security in Sub-Saharan Africa. Annu. Rev. Phytopathol. 56, 381–403.

Dai, M.M., Liu, R., Jiang, H., Zhang, X.P., Song, W.W., Zhang, J., Liang, C., Zhao, H.H., Shi, Q.Q., 2023. Volatile organic compounds of *Bacillus pumilus* strain S1-10 exhibit fumigant activity against *Meloidogyne incognita*. Plant Dis. 107, 3057–3063.

Dinardo-Miranda, L.L., Miranda, I.D., Silva, H.D.S.D., Fracasso, J., 2022. Biological control of phytoparasitic nematodes in sugarcane fields. Pesq. Agrop. Trop. 52, 1–7.

Dong, L., Zhang, K., 2006. Microbial control of plant-parasitic nematodes: a five-party interaction. Plant and Soil 288, 31–45.

Edwards, C.A., Arancon, N.Q., Emerson, E., Pulliam, R., 2007. Suppressing plant parasitic nematodes and arthropod pests with vermicompost teas. BioCycle 48, 38–39.

El-Nagdi, W.M.A., Youssef, M.M.A., 2004. Soaking faba bean seed in some bio-agents as prophylactic treatment for controlling *Meloidogyne incognita* root-knot nematode infection. J. Pest. Sci. 77, 75–78.

Fan, X., Ge, A.H., Qi, S., Guan, Y., Wang, R., Yu, N., Wang, E., 2025. Root exudates and microbial metabolites: signals and nutrients in plant-microbe interactions. Sci. China Life Sci. 68, 2290–2302.

Ferrara, F.I.D.S., Oliveira, Z.M., Gonzales, H.H.S., Floh, E.I.S., Barbosa, H.R., 2012. Endophytic and rhizospheric enterobacteria isolated from sugar cane have different potentials for producing plant growth-promoting substances. Plant and Soil 353, 409–417.

Gebrehana, Z.G., Gebremikael, M.T., Beyene, S., Wesemael, W.M.L., De Neve, S., 2025. Suppression potential of selected vermicomposts against root-knot nematode (*Meloidogyne incognita*) under in vitro, pot, and field conditions. Front. Plant Sci. 16, 1–18.

Gowda, A.P.A., Pankaj, D., Singh, D., Sowmya, R., 2022. Nematicidal potential of plant growth-promoting rhizobacteria against *Meloidogyne incognita* infesting tomato under protected cultivation. Egypt. J. Biol. Pest Control 32, 1–13.

Hackenberg, C., Muehlkchen, A., Forge, T., Vrain, T., 2000. *Pseudomonas chlororaphis* strain Sm3, bacterial antagonist of *Pratylenchus penetrans*. J. Nematol. 32, 183.

Jenkins, W., 1964. A rapid centrifugal flotation technique for extracting nematodes from soil. Plant Dis. Rep. 48, 692.

Lazarovits, G., Tenuta, M., Conn, K.L., 2001. Organic amendments as a disease control strategy for soilborne diseases of high-value agricultural crops. Australas. Plant Pathol. 30, 111–117.

Lee, J.H., Ma, K.C., Ko, S.J., Kang, B.R., Kim, I.S., Kim, Y.C., 2011. Nematicidal activity of a nonpathogenic biocontrol bacterium, *Pseudomonas chlororaphis* O6. Curr. Microbiol. 62, 746–751.

Li, Z.G., Luo, Y.M., Teng, Y., 2008. Research Method of Soil and Environmental Microorganisms. Science press, Beijing.

Li, J., Zou, C., Xu, J., Ji, X., Niu, X., Yang, J., Huang, X., Zhang, K.Q., 2015. Molecular mechanisms of nematode - nematophagous microbe interactions: basis for biological control of plant-parasitic nematodes. Annu. Rev. Phytopathol. 53, 67–95.

Liang, J.G., Sun, S., Ji, J., Wu, H.Y., Meng, F., Zhang, M.R., Zheng, X.B., Wu, C.X., Zhang, Z.G., 2014. Comparison of the rhizosphere bacterial communities of zigongdongdou soybean and a high-methionine transgenic line of this cultivar. PLoS One 9, 1–10.

Liang, C., Yang, D.W., Dong, F.Y., Shang, J.X., Niu., X.G., Zhang, G.X., Yang, L.J., Wang, Y.H., 2024. Biocontrol potential of bacteria isolated from vermicompost against *Meloidogyne incognita* on tomato and cucumber crops. Horticulture 10, 1–13.

Liu, D.W., Han, W.H., Zhang, Y.J., Wu, F.Z., Liu, T., 2017. Evaluation of control effect of vermicompost and its extracts on tomato root-knot nematode. Chin. J. Biol. Control 33 (5), 686–691 (Chinese).

Lu, P., Shi, H.L., Tao, J.M., Jin, J.J., Wang, S.J., Zheng, Q.X., Liu, P.P., Xiang, B.K., Chen, Q.S., Xu, Y.L., Li, Z.F., Tan, J., Cao, P.J., 2023. Metagenomic insights into the changes in the rhizosphere microbial community caused by the root-knot nematode *Meloidogyne incognita* in tobacco. Environ. Res. 216, 1–10.

Meyer, S.L.F., 2003. United States Department of Agriculture - Agricultural Research Service research programs on microbes for management of plant - parasitic nematodes. Pest Manag. Sci. 59, 665–670.

Mishra, S., Wang, K.H., Sipes, B.S., Tian, M., 2018. Induction of host-plant resistance in cucumber by vermicompost tea against root-knot nematode. Nematropica 48, 164–171.

Nakasone, A.K., Bettiol, W., De Souza, R.M., 1999. The effect of water extracts of organic matter on plant pathogens. Summa Phytopathol. 25, 330–335.

Oka, Y., 2010. Mechanisms of nematode suppression by organic soil amendments-a review. Appl. Soil Ecol. 44, 101–115.

Orlikowski, L.B., 1999. Vermicompost extract in the control of some soil-borne pathogens. Environ. Sci. Agric. Food Sci. 64, 405–410.

Rehman, S.U., De Castro, F., Aprile, A., Benedetti, M., Fanizzi, F.P., 2023. Vermicompost: enhancing plant growth and combating abiotic and biotic stress. Agronomy 13, 1134.

Rostami, M., Karegar, A., Taghavi, S.M., 2021. Biocontrol potential of bacterial isolates from vermicompost and earthworm against the root-knot nematode *Meloidogyne javanica* infecting tomato plants. Egypt. J. Biol. Pest Control 31, 1–11.

Settu, V., Annaiyan, S., Mannu, J., 2023. Revealing the genetic arsenal of *Bacillus firmus* TNAU1: unleashing nematicidal and plant growth promotion traits. Physiol. Mol. Plant Pathol. 129, 1–11.

Siddiqui, Z.A., 2004. Effects of plant growth promoting bacteria and composed organic fertilizers on the reproduction of *Meloidogyne incognita* and tomato growth. Bioresour. Technol. 95, 223–227.

Siddiqui, I.A., Shaukat, S.S., 2003. Suppression of root-knot disease by *Pseudomonas fluorescens* CHA0 in tomato: importance of bacterial secondary metabolite, 2, 4-diacetylpholoroglucinol. Soil Biol. Biochem. 35, 1615–1623.

Siddiqui, I.A., Haas, D., Heeb, S., 2005. Extracellular protease of *Pseudomonas fluorescens* CHA0, a biocontrol factor with activity against the root-knot nematode *Meloidogyne incognita*. Appl. Environ. Microbiol. 71, 5646–5649.

Sikora, R.A., Schäfer, K., Dababat, A.A., 2007. Modes of action associated with microbially induced in planta suppression of plant-parasitic nematodes. Australas. Plant Pathol. 36, 124–134.

Song, X.H., Li, Y.K., Song .L., Y, Zheng, Y.K., Hou, Z.L., Yu, D.P., 2021. Characteristic analysis of microbial community in vermicompost. J. Anhui Univ. (Nat. Sci. Ed.) 45, 92–97 (Chinese).

Sun, X., Xu, Z., Xie, J., Hesselberg-Thomsen, V., Tan, T., Zheng, D., Strube, M.L., Drago, A., Shen, Q., Zhang, R., 2021. *Bacillus velezensis* stimulates resident rhizosphere *Pseudomonas stutzeri* for plant health through metabolic interactions. ISME J. 16 (3), 774–787.

Sun, J.C., Li, X.J., Qu, Z., Wang, H.R., Cheng, Y., Dong, S.J., Zhao, H., 2023. Comparative proteomic analysis reveals novel insights into the continuous cropping induced response in Scrophularia ningpoensis. J. Sci. Food Agric. 103, 1832–1845.

Szczech, M.M., 1999. Suppression of vermicompost against Fusarium wilt of tomato. J. Phytopathol. 147, 155–161.

Szczech, M., Smolińska, U., 2001. Comparison of suppressiveness of vermicomposts produced from animal manures and sewage sludge against *Phytophthora nicotianae* Breda de Haan var. *nicotianae*. J. Phytopathol. 149, 77–82.

Tao, C., Li, R., Xiong, W., Shen, Z., Kowalchuk, G.A., 2020. Bio-organic fertilizers stimulate indigenous soil *Pseudomonas* populations to enhance plant disease suppression. Microbiome 8, 1–14.

Tian, B., Yang, J., Zhang, K.Q., 2007. Bacteria used in the biological control of plant-parasitic nematodes: populations, mechanisms of action, and future prospects. FEMS Microbiol. Ecol. 61, 197–213.

Tian, B.Y., Cao, Y., Zhang, K.Q., 2015. Metagenomic insights into communities, functions of endophytes and their associates with infection by root-knot nematode, *Meloidogyne incognita*, in tomato roots. Sci. Rep. 5, 1–15.

Wahla, V., Maheshwari, D.K., Bajpai, V.K., 2012. Nematicidal *fluorescent Pseudomonads* for the in vitro and in vivo suppression of root knot (*Meloidogyne incognita*) of *Capsicum annuum* L. Pest Manag. Sci. 68, 1148–1155.

Wang, J., Wen, C., Liu, W., Dong, D., Qiu, J., Liu, T., 2016. Screening and identification of biocontrol bacterial against root-knot nematode. North. Hortic. 9, 121–124.

Wang, J., Qiu, M., Shen, Z., Chen, L., Ge, Y., 2025. Microbial modulation of soil pH: a self-benefiting mechanism exemplified by *Bacillus*. Soil Biol. Biochem. 210, 1–10.

Xia, L., Miao, Y., Cao, A., Liu, Y., Liu, Z., Sun, X., Xue, Y., Xu, Z., Xun, W., Shen, Q., 2022. Biosynthetic gene cluster profiling predicts the positive association between antagonism and phylogeny in *Bacillus*. Nat. Commun. 13, 1–11.

Xiao, Z., Liu, M., Jiang, L., Chen, X., Feng, H., 2016. Vermicompost increases defense against root-knot nematode (*Meloidogyne incognita*) in tomato plants. Appl. Soil Ecol. 105, 177–186.

Yang, H.W., Li, J., Xiao, Y.H., Gu, Y.B., Liu, H.W., Liang, Y.L., Liu, X.D., Hu, J., Meng, D. L., Yin, H.Q., 2017. An integrated insight into the relationship between soil microbial community and tobacco bacterial wilt disease. Front. Microbiol. 8, 1–11.

Yang, F., Jiang, H.Y., Liang, S., Chang, G.Z., Ma, K., Niu, L.L., Mi, G.Q., Tang, Y.L., Tian, B.M., Shi, X.J., 2023. High-throughput sequencing reveals the effect of the south root-knot nematode on cucumber rhizosphere soil microbial community. Agronomy 13, 1–14.

Ye, L., WANG, J.Y., Liu, X.F., Guan, Q., Dou, N.X., Li, J., Zhang, Q., Gao, Y.M., Wang, M., Li, J.S., 2022. Nematicidal activity of volatile organic compounds produced by *Bacillus altitudinis* AMCC 1040 against *Meloidogyne incognita*. Arch. Microbiol. 204, 521.

Yin, N., Zhao, J.L., Liu, R., Mao, Z., 2021. Biocontrol efficacy of *Bacillus cereus* strain Bc-cm103 against *Meloidogyne incognita*. Plant Dis. 105, 2061–2070.

Yu, G., Ge, X., Li, W., Ji, L., Yang, S., 2024. Interspecific cross-talk: the catalyst driving microbial biosynthesis of secondary metabolites. Biotechnol. Adv. 76, 1–21.

Zhai, Y., Shao, Z., Cai, M., Zheng, L., Li, G., Huang, D., Cheng, W., Thomashow, L.S., Weller, D.M., Yu, Z., 2018. Multiple modes of nematode control by volatiles of *Pseudomonas putida* 1A00316 from Antarctic soil against *Meloidogyne incognita*. Front. Microbiol. 9, 253.

Zhang, L.F., Gui, T.R., Dai, D.Q., Wei, X.M., Yin, S., Hu, H.L., 2020. Comparative analysis bacterial community diversity of potato rhizosphere soil caused by root-knot nematodes using high-throughput sequencing. Mol. Plant Breed. 1–13 (Chinese).

Zhao, F.Y., Zhang, Y.Y., Dong, W.G., Zhang, Y.Q., Zhang, G.X., Sun, Z.P., Yang, L.J., 2019. Vermicompost can suppress *fusarium oxysporum f. sp. lycopersici* via generation of beneficial bacteria in a long-term tomato monoculture soil. Plant Soil 440, 491–505.

Zhou, Y., Coventry, D.R., Gupta, V.V.S.R., Fuentes, D., Merchant, A., Kaiser, B.N., Li, J., Wei, Y., Liu, H., Wang, Y., Gan, S., Denton, M.D., 2020. The preceding root system drives the composition and function of the rhizosphere microbiome. Genome Biol. 21, 1–19.

Zhou, D.M., Feng, H., Schuelke, T., De Santiago, A., Zhang, Q.M., Zhang, J.F., Luo, C.P., Wei, L., 2019a. Rhizosphere microbiomes from root knot nematode non - infested plants suppress nematode infection. Microb. Ecol. 78, 470–481.

Zhou, G., Qiu, X., Chen, L., Zhang, C., Ma, D., Zhang, J., 2019b. Succession of organics metabolic function of bacterial community in response to addition of earthworm casts and zeolite in maize straw composting. Bioresour. Technol. 280, 229–238.

Zuckerman, B., M., Dicklow, M.B., Coles, G.C., Garcia-E, R., Marban-Mendoza, N., 1989. Suppression of plant parasitic nematodes in the chinampa agricultural soils. J. Chem. Ecol. 15, 1947–1955.

Zuhair, R., Moustafa, Y.T.A., Mustafa, N.S.A., El-Dahshouri, M., F., Zhang, L., Ageba, M. F., 2022. Efficacy of amended vermicompost for bio-control of root knot nematode (RKN) *Meloidogyne incognita* infesting tomato in Egypt. Environ. Technol. Innov. 27, 1–8.